

CPSC 470/570 Problem Set 2

Answer Sheet

Instructions:

1. You may enter your answers directly into this PDF form or using an annotation tool. For the sake of the environment and our autograder, please avoid printing & re-scanning this sheet.
2. Enter your answers into the **yellow boxes**. Anything written outside of a yellow box will be ignored.
3. Upload your work to Gradescope as a separate document. Please do **not** show your work on this sheet.
4. Round numerical answers to two decimal places.

Q1a)

$U(A1) =$

$U(A2) =$

$U(B1) =$

$U(B2) =$

Q1b)

Q1c)

$U(A1) =$

$U(A2) =$

$U(B1) =$

$U(B2) =$

Q1d)

Q2a)

$T(A1, \uparrow, A1) =$ 	$T(A1, \uparrow, B1) =$ 	$T(A1, \uparrow, A2) =$ 	$T(A1, \uparrow, B2) =$
$T(A1, \downarrow, A1) =$ 	$T(A1, \downarrow, B1) =$ 	$T(A1, \downarrow, A2) =$ 	$T(A1, \downarrow, B2) =$
$T(A1, \Leftarrow, A1) =$ 	$T(A1, \Leftarrow, B1) =$ 	$T(A1, \Leftarrow, A2) =$ 	$T(A1, \Leftarrow, B2) =$
$T(A1, \Rightarrow, A1) =$ 	$T(A1, \Rightarrow, B1) =$ 	$T(A1, \Rightarrow, A2) =$ 	$T(A1, \Rightarrow, B2) =$

$T(A2, \uparrow, A1) =$ 	$T(A2, \uparrow, B1) =$ 	$T(A2, \uparrow, A2) =$ 	$T(A2, \uparrow, B2) =$
$T(A2, \downarrow, A1) =$ 	$T(A2, \downarrow, B1) =$ 	$T(A2, \downarrow, A2) =$ 	$T(A2, \downarrow, B2) =$
$T(A2, \Leftarrow, A1) =$ 	$T(A2, \Leftarrow, B1) =$ 	$T(A2, \Leftarrow, A2) =$ 	$T(A2, \Leftarrow, B2) =$
$T(A2, \Rightarrow, A1) =$ 	$T(A2, \Rightarrow, B1) =$ 	$T(A2, \Rightarrow, A2) =$ 	$T(A2, \Rightarrow, B2) =$

Q2b)

$$Q^*(A1, \hat{\cup}) =$$

$$Q^*(A1, \hat{\cap}) =$$

$$Q^*(A1, \hat{\Delta}) =$$

$$Q^*(A1, \hat{\ominus}) =$$

$$Q^*(A2, \hat{\cup}) =$$

$$Q^*(A2, \hat{\cap}) =$$

$$Q^*(A2, \hat{\Delta}) =$$

$$Q^*(A2, \hat{\ominus}) =$$

Q2c)

$$Q^*(A1, \hat{\cup}) =$$

$$Q^*(A1, \hat{\cap}) =$$

$$Q^*(A1, \hat{\Delta}) =$$

$$Q^*(A1, \hat{\ominus}) =$$

$$Q^*(A2, \hat{\cup}) =$$

$$Q^*(A2, \hat{\cap}) =$$

$$Q^*(A2, \hat{\Delta}) =$$

$$Q^*(A2, \hat{\ominus}) =$$

Q3a)

Indicate the value of each state by writing it in the corresponding grid cell.

 $V_1(s):$

E				
D				
C				
B				
A				
	1	2	3	4

$V_3(s)$:

E					
D					
C					
B					
A					
	1	2	3	4	5

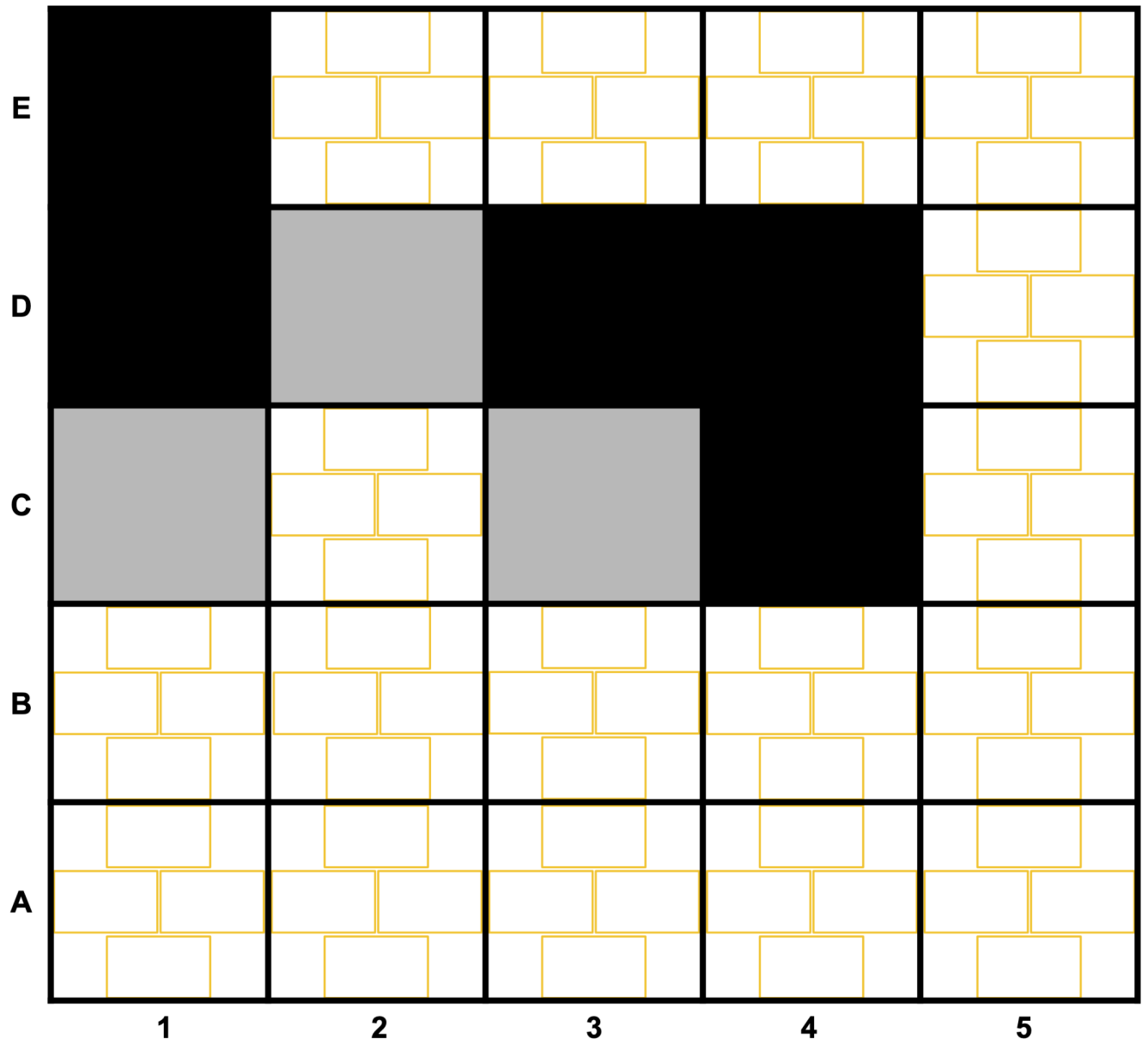
$V_5(s)$:

E					
D					
C					
B					
A					
	1	2	3	4	5

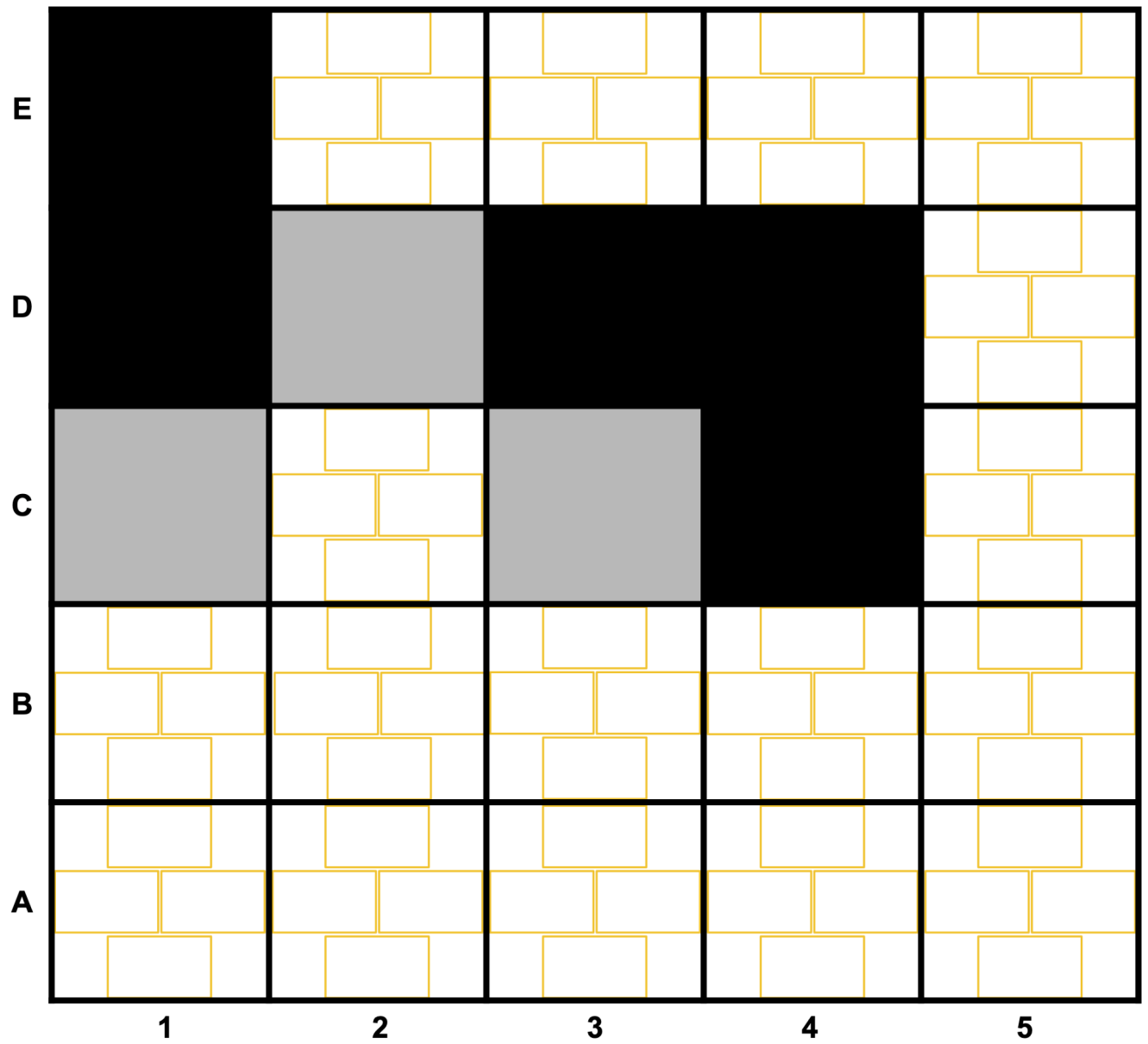
Q3b)

For each state, indicate the Q-value for each action like this:

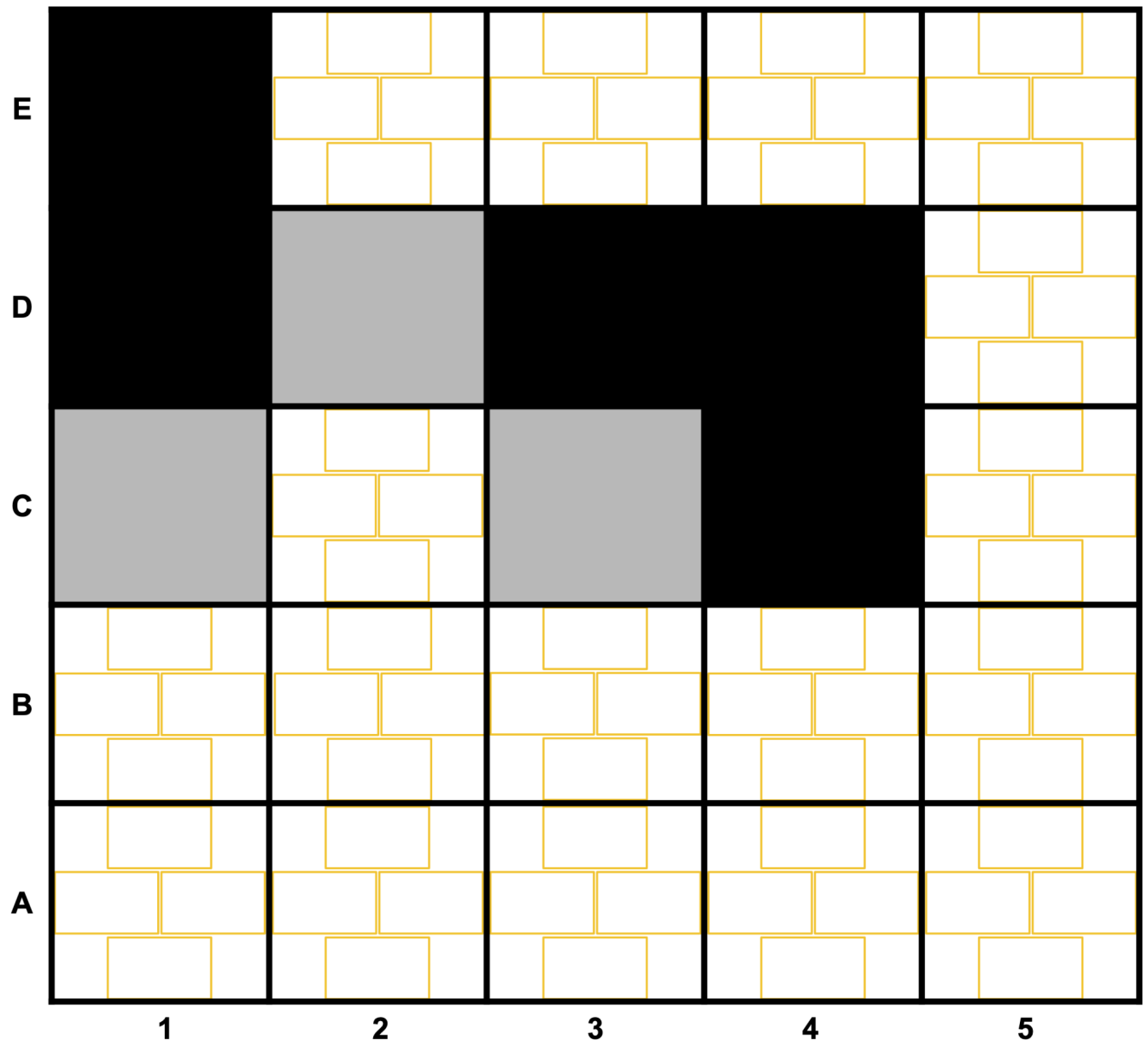
	0.23	
0.58		0.41
	0.09	

 $Q_1(s,a):$ 

$Q_3(s,a):$

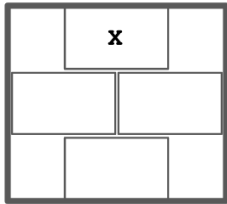


$Q_5(s,a):$



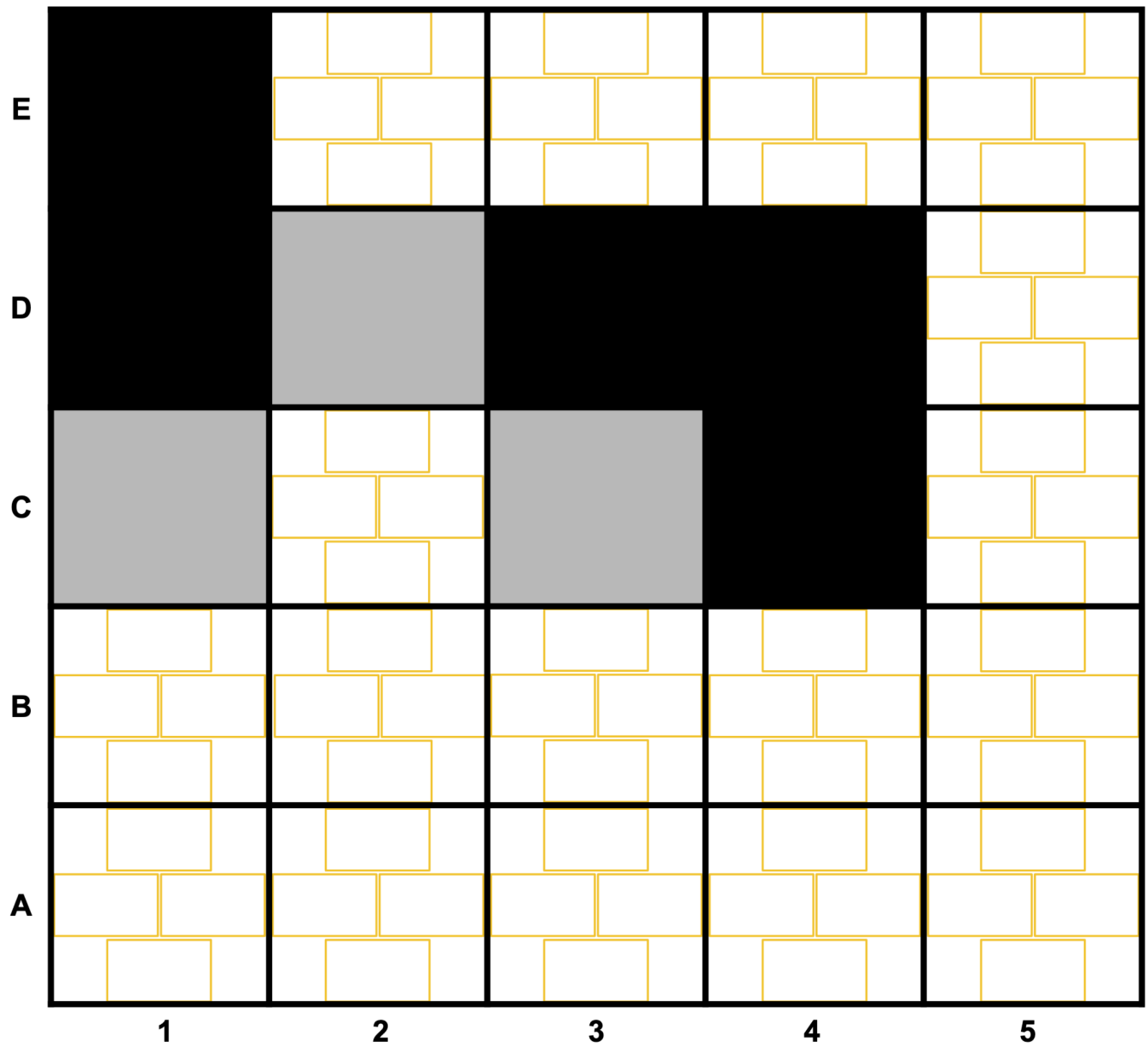
Q3c)

For each state, indicate which action(s) the policy returns like this:

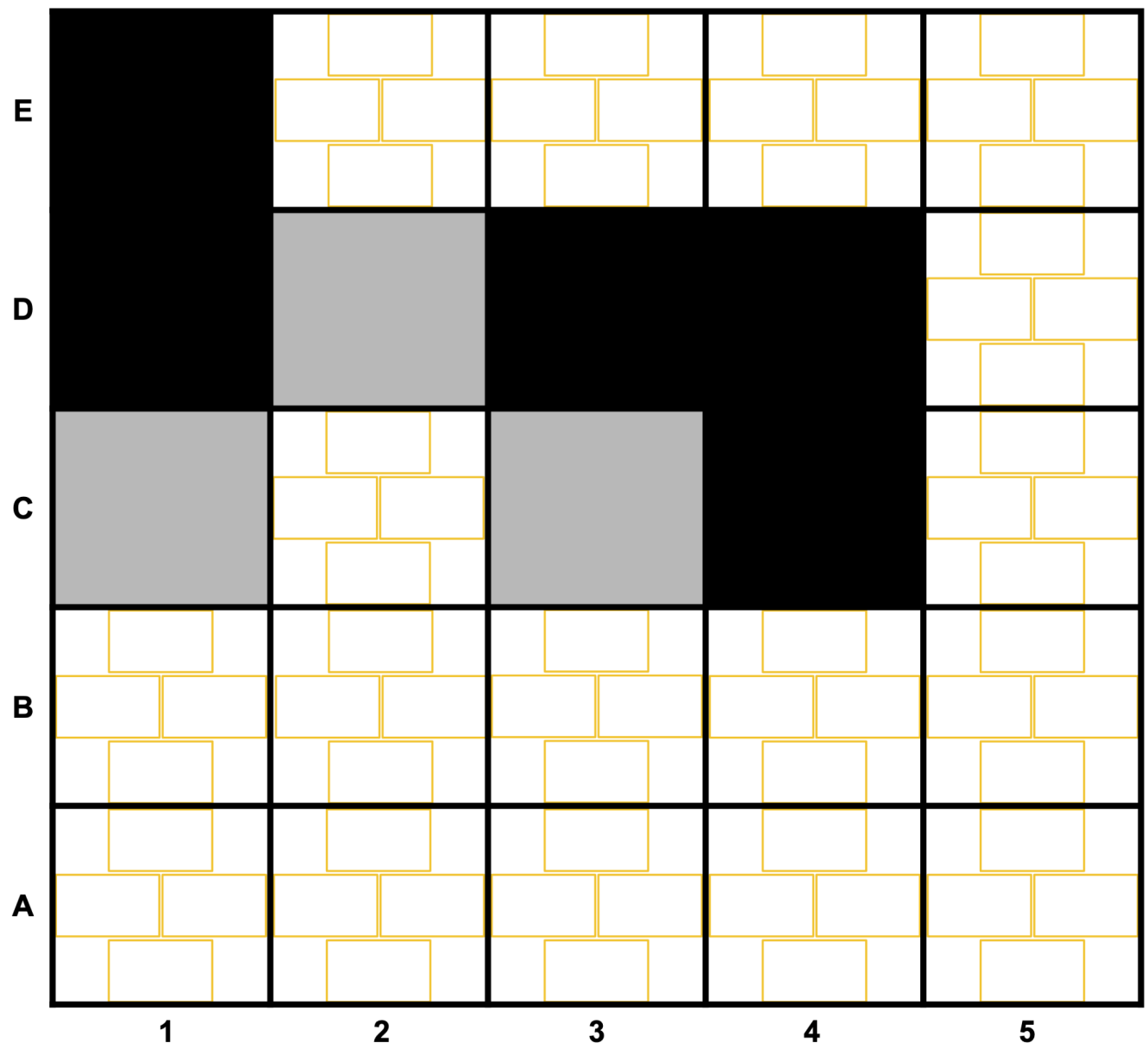


If multiple actions are equivalent, place an 'X' on all equivalent actions.

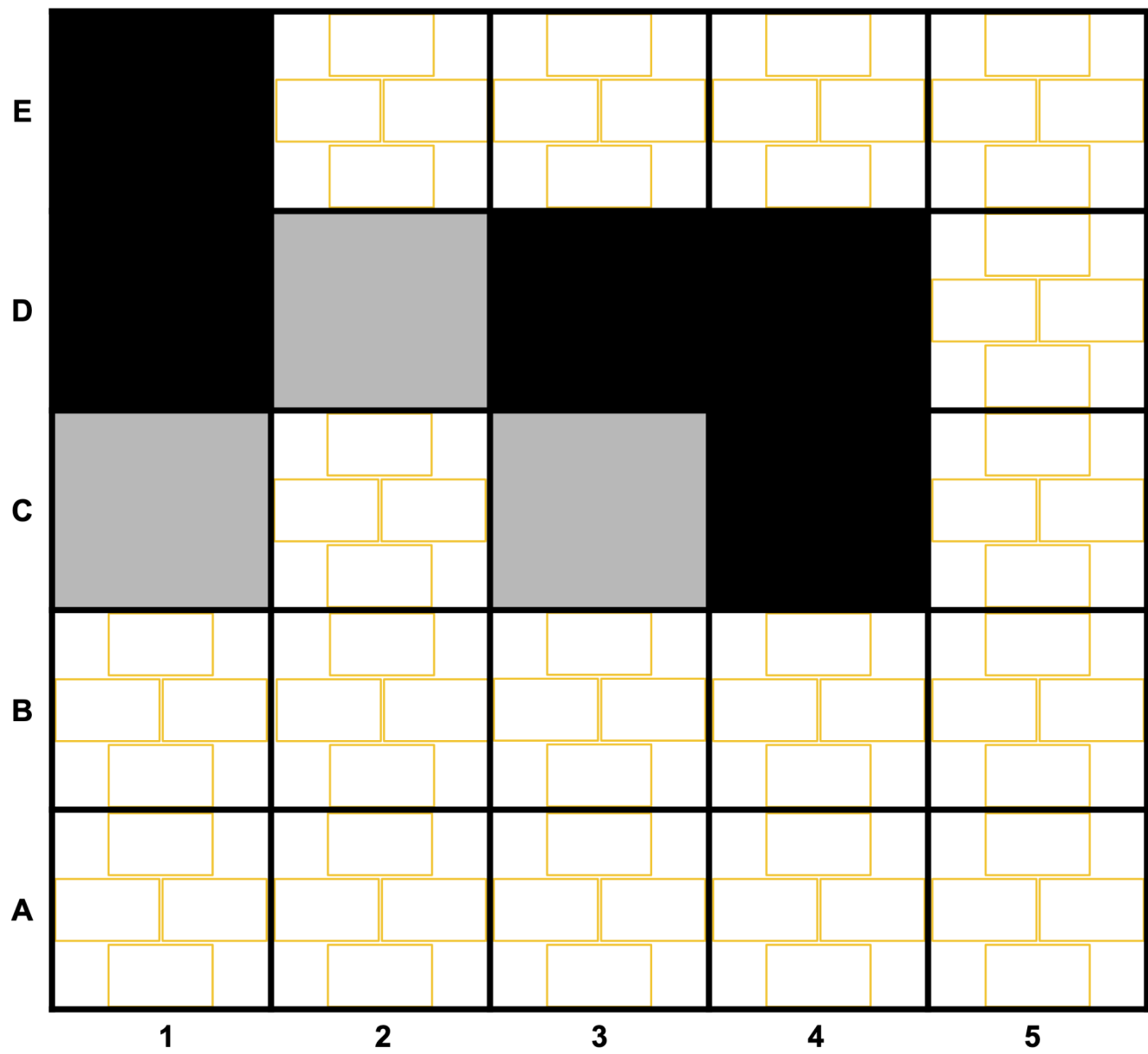
$\pi_1(s)$:



$\pi_3(s)$:



$\pi_5(s)$:



Q3d)

Indicate the value of each state by writing it in the corresponding grid cell.

$$V^{\pi_1}(s)$$

E				
D				
C				
B				
A				
	1	2	3	4

$V_3^\pi(s)$

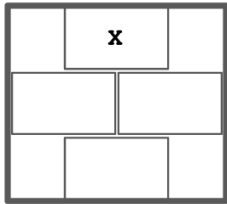
E					
D					
C					
B					
A					
	1	2	3	4	5

$V_5^\pi(s)$

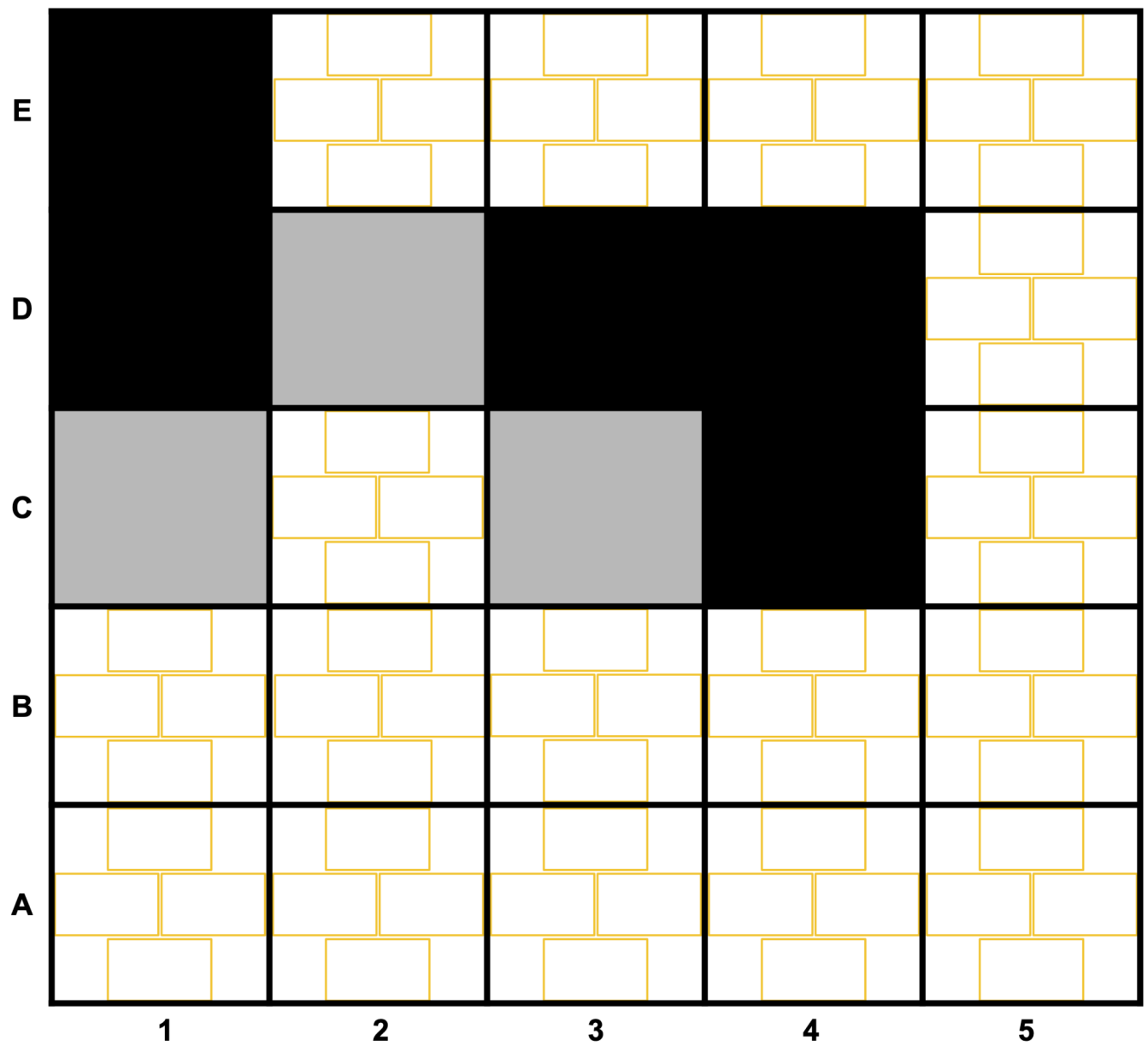
E					
D					
C					
B					
A					
	1	2	3	4	5

Q3e)

For each state, indicate which action(s) the policy returns like this:



If multiple actions are equivalent, place an 'X' on all equivalent actions.



Q4a)

$v^{\pi^0}(1) =$ 	$v^{\pi^0}(2) =$ 	$v^{\pi^0}(3) =$ 	$v^{\pi^0}(4) =$ 	$v^{\pi^0}(5) =$
$v^{\pi^1}(1) =$ 	$v^{\pi^1}(2) =$ 	$v^{\pi^1}(3) =$ 	$v^{\pi^1}(4) =$ 	$v^{\pi^1}(5) =$
$v^{\pi^2}(1) =$ 	$v^{\pi^2}(2) =$ 	$v^{\pi^2}(3) =$ 	$v^{\pi^2}(4) =$ 	$v^{\pi^2}(5) =$
$v^*(1) =$ 	$v^*(2) =$ 	$v^*(3) =$ 	$v^*(4) =$ 	$v^*(5) =$

Q4b)

$v^{\pi^0}(1) =$ 	$v^{\pi^0}(2) =$ 	$v^{\pi^0}(3) =$ 	$v^{\pi^0}(4) =$ 	$v^{\pi^0}(5) =$
$v^{\pi^1}(1) =$ 	$v^{\pi^1}(2) =$ 	$v^{\pi^1}(3) =$ 	$v^{\pi^1}(4) =$ 	$v^{\pi^1}(5) =$
$v^{\pi^2}(1) =$ 	$v^{\pi^2}(2) =$ 	$v^{\pi^2}(3) =$ 	$v^{\pi^2}(4) =$ 	$v^{\pi^2}(5) =$
$v^*(1) =$ 	$v^*(2) =$ 	$v^*(3) =$ 	$v^*(4) =$ 	$v^*(5) =$

Q4c)

Q5a)

C					
B					
A					
	1	2	3	4	5

Q5b)

C					
B					
A					
	1	2	3	4	5

Q5c)

C				
B				
A				
	1	2	3	4
		5		

Q5d)

C					
B					
A					
	1	2	3	4	5

Q5e)

C					
B					
A					
	1	2	3	4	5

Q5f)

C					
B					
A					
	1	2	3	4	5